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Code and Data Availability

The problem sets in full are included in Appendices A and B. We highly encourage readers to test their LLM assistants on these problems, especially the games of 24. The complete data generated in this work can be found here: <https://osf.io/fncyu/>.

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A Prompts Used in Study 1

Prompt 1: Hi. I will give you some sets of mathematical problems. Solve them as best as you can. For any fractions, provide 3 digits after the decimal point. You are welcome to provide the reasonings, but please remember to summarize all answers in a list following the order of the problems presented. Do you understand?

(After an affirmative response from the agent)

Prompt 2: Here is the first set of problems. Simply calculate these values.

1. $1 + 2 - 3 * 4/5$
2. $2 + 2 + 4 - 2$
3. $(3 + 4 * 3)/5$
4. $1/13$
5. $13 * 7/(169 * 25)$
6. $(13 + 4)(13 - 4)$
7. $1/1.2 + 1/6$
8. $(1/13 - 1/11) * 999999$
9. $87/29 + 2$
10. 49π
11. $\pi/6$
12. $e + \pi$
13. $79/13$
14. $1 + 2 + 3 + 4 + 5 + 6 + 7$
15. $1 - 2 + 3 - 4 + 5 - 6 + 7 - 8$
16. $1 - 1 + 1 - 1 + 1$
17. $55 - 1 - 2 - 3 - 4 - 5 - 6 - 7 - 8 - 9$
18. $1 + 1/2 + 1/3 + 1/4 + 1/5 + 1/6 + 1/7$
19. $1/10 + 1/100 + 1/1000 + 1/10000$
20. $11 * 101/9999$
21. $1/(11 * 101 * 73 * 137)$
22. $(11 * 13 * 17 * 19)/(22 * 39 * 68 * 95)$
23. $1/(3/10) - 20/6$
24. $1/(2 * 12 - 3 * (73 - 69))$
25. $1/(2 * 12 - 2 * 3 * (73 - 69))$

Prompt 3: Okay. Now, calculate these values.

1. 0^0
2. 1^0
3. 2^3
4. 2^{-5}
5. $400^{-.5}$
6. $400^{0.23}$
7. $237^{3.7}$
8. $500^{2/7}$